

PRISMA 2020 Flow Diagram

Identification

Records identified from databases

(n = 1,687)

PubMed 286 | Embase 375 | Scopus 398

Web of Science 319 | CENTRAL 309

Duplicate records removed

(n = 562)

Records after deduplication

(n = 1,125)

Screening

Records screened (title / abstract)

(n = 1,125)

Records excluded (n = 874)

No NMN/NR intervention (n = 770)

Not an RCT (n = 39)

Review / protocol (n = 32)

Animal / in vitro (n = 27)

Pediatric / T1DM (n = 6)

Reports sought for retrieval

(n = 251)

Reports not retrieved (n = 0)

Eligibility

Reports assessed for eligibility

(full-text review)

(n = 251)

Reports excluded (n = 217)

Title/abstract screening (n = 133)

Not RCT / no placebo arm

Non-metabolic primary outcome

Non-human / in vitro

Re-screening of uncertain (n = 65)

Protocol / commentary only

Duplicate cohort data

Manual full-text review (n = 19)

No numeric endpoint data

Combination supplement

Population outside PICO

Reports meeting inclusion criteria

(n = 34; NMN = 14, NR = 20)

Reports excluded from qualitative synthesis (n = 19)

Trial registration only (n = 9)

No extractable data / image PDF (n = 3)

Combination supplement (n = 2)

Secondary/duplicate publication (n = 5)

Included

Studies included in qualitative synthesis (n = 15)

NMN = 5 | NR = 10

Studies excluded from quantitative NMA (n = 7)

No metabolic endpoint data (n = 5)

High RoB + insufficient data (n = 2)

Studies included in quantitative synthesis — NMA (n = 8)

NMN = 4 | NR = 4

21 outcomes | 73 data points